

Cheap Metrics First, Replanning Diagnostics Second: Offline Checkpoint Screening for Robot Imitation Learning

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Abstract—Offline checkpoint selection in robot imitation learning is usually treated as loss minimization, although deployment changes the policy’s future inputs after every action. We present an empirical study of what is gained by moving from objective loss and Action L1 to open-loop regularity, teacher-forced replanning, learned-model feedback, and simulator feedback. Across 1,365 low-dimensional robomimic checkpoints, PLV-1 smoothness and Action L1 have similar task-balanced rank correlation (0.434 and 0.418), while Action L1 has lower Regret@30 (0.033 versus 0.045) and dynamics-free open-loop smoothness is lowest (0.019). The pooled result, however, hides a policy-family reversal: PLV-1 is strongest among the reported screens on Diffusion Policy runs, whereas cheap metrics dominate on BC-RNN. Execution matching reveals a second boundary. In a hard-Diffusion-Policy discovery pool, boundary inconsistency reaches $\rho = 0.730$ versus 0.512 for Action L1, showing that temporal disagreement can expose behavior missed by pointwise action error. Matched and external controls do not support turning that signal into a universal learned-model reranker; simulator-induced inconsistency is adversely ordered in the same-start audit ($\rho = -0.212$). The study therefore separates three questions often conflated in checkpoint evaluation: ranking versus selection, temporal diagnosis versus global scoring, and transition accuracy versus decision quality. Cheap action-space metrics remain robust screens, while replanning diagnostics identify policy-family-dependent disagreement and when rollout evidence is still required.

Index Terms—imitation learning, robot learning, checkpoint selection, policy evaluation, action chunking, learned dynamics

I. INTRODUCTION

Imitation-learning training can produce tens of checkpoints for each task, seed, and policy family. Evaluating all of them with full simulator or robot episodes is expensive, so selection often falls back to the training objective or an action error on held-out demonstrations. The selected checkpoint nevertheless determines the policy that is reported or placed on hardware, making the choice an offline policy-evaluation problem rather than a bookkeeping detail.

The source of the mismatch is simple. Open-loop validation queries a policy at an expert state, $a_t = \pi_\theta(x_t^{\text{expert}})$, and compares the result with a recorded action. Deployment instead changes the input distribution after every action. Contact timing, precision alignment, mode switching, and recovery can

therefore depend on a sequence of policy-induced observations even when the first action looks reasonable. Our motivating manipulator experiments exposed exactly this pattern: checkpoints with similar logged errors could behave differently near contact. We use that experience only as motivation; all empirical claims in this paper are simulation-based.

A learned transition model does not automatically solve this mismatch. Model error may accumulate, action regularity may explain an apparent gain, and accurate transition prediction need not preserve policy order. Rather than treating model-based validation as intrinsically stronger, we organize checkpoint evidence into successive stages: action error, dynamics-free temporal regularity, teacher-forced replanning, model-induced feedback, and simulator feedback. Each stage is asked to justify the extra claim and cost it introduces.

This study finds structured heterogeneity rather than a universal winner. Cheap metrics are strong first-pass selectors, but their apparent equivalence in pooled results conceals a reversal between BC-RNN and Diffusion Policy. Execution-matched replanning exposes an additional hard-task signal, while matched controls show that this signal diagnoses temporal disagreement more reliably than it authorizes automatic reranking. The resulting contribution is both methodological and practical: it identifies which validation question each metric can answer, and where offline evidence should give way to rollout.

Our contributions are:

- 1) We organize checkpoint evaluation as an evidence hierarchy separating cheap screens from teacher-forced, model-induced, and simulator-induced temporal diagnostics.
- 2) We evaluate 1,365 checkpoints with task-balanced ranking, selection regret, finite-rollout reliability, and margin-aware violations.
- 3) We show that pooled averages hide a policy-family reversal and a gap between global ranking and top-of-pool selection.
- 4) We isolate execution-matched boundary and overlap inconsistency, then use retained, simulator, and external controls to separate conditional temporal diagnosis from unsupported universal reranking.

II. RELATED WORK

a) *Imitation-learning checkpoint selection*: Behavior cloning is vulnerable to covariate shift because training states come from an expert while deployment states are induced by the learner [1], [2]. Robomimic documents that the policy objective and rollout success need not agree [5], and reward-free hyperparameter selection over imitation agents shows that proxy choice changes the selected policy [13]. We study the narrower but common problem of choosing among checkpoints from a fixed robot-imitation run.

b) *Offline and model-based policy evaluation*: Offline policy selection and OPE ask how logged data can rank policies without executing all candidates [12], [14], [15]. Decision-targeted models emphasize that one-step prediction accuracy need not preserve the downstream policy decision [20]. Large visual world-model evaluators estimate task success for robot policies [9], [10], [11]. PLV occupies a lighter, task-local point on this spectrum: it screens many intermediate checkpoints and reports decomposed diagnostics rather than estimating a universal value. SIMPLER’s Mean Maximum Rank Violation (MMRV) weights a ranking error by the true performance margin [22]; we report it alongside correlation so harmless near-ties do not count like large success reversals.

c) *Failure detection and distribution shift*: Runtime monitors, observation OOD tests, and generative-policy uncertainty can identify unsafe or unfamiliar execution states [16], [17], [18]. PLV differs in timing and target: it is applied offline before deployment, and its target is the relative ordering or escalation priority of checkpoints. The present experiments also show why calibrated transition uncertainty alone is insufficient for that decision.

d) *Action chunking and replanning stability*: Diffusion Policy and ACT predict action sequences rather than a single primitive command [4], [3]. Their deployed behavior depends on observation horizon, execution stride, and when the policy is queried again. Adaptive and masked chunk-execution methods likewise treat replanning consistency as a control issue [19], [21]. Our execution-matched audit uses the policy’s actual eight-action stride and measures disagreement at the point where an old plan is replaced by a new one.

III. PROBLEM SETUP

Let $\Pi = \{\pi_{\theta_1}, \dots, \pi_{\theta_N}\}$ be checkpoints from one training run and let

$$S_i = \mathbb{E}[\mathbb{1}(\text{task success} \mid \pi_{\theta_i})] \quad (1)$$

denote the unknown deployed success probability. A validation metric produces $M_i = M(\pi_{\theta_i}; \mathcal{D}_{\text{val}})$. The selection question is whether the ordering induced by M_i preserves the ordering of S_i .

Objective validation loss and Action L1 are not interchangeable. BC-RNN is trained by a mixture-density likelihood but Action L1 compares its decoded action with the expert action. Diffusion Policy validates a denoising objective while Action L1 compares sampled final actions. ACT combines chunk reconstruction with latent regularization while Action

L1 compares the predicted and expert chunks. Thus Action L1 is a common action-space comparator across otherwise incompatible objectives.

Our primary endpoints are direction-oriented Spearman correlation and Regret@30. Costs are negated before correlation, so positive values always rank better checkpoints higher. For a retained fraction $q = 0.30$,

$$\text{Regret}@q = S^* - \max_{i \in \text{Top}_q(M)} S_i, \quad (2)$$

where $S^* = \max_i S_i$. We write Regret@30 for this $q = 0.30$ budget; R@10% and R@1 are separate endpoints reported in the supplement. Kendall’s τ_b , tighter budgets, residual association, and pairwise ordering are secondary. Checkpoint-level correlations describe ordering within a pool; they are not independent replicates. We first compute $\rho_{t,r}$ within each task-specific training run, average runs within task, and then weight tasks equally. Task-clustered bootstrap intervals resample tasks and their runs together.

IV. DIAGNOSTIC HIERARCHY

A. Cheap action-space controls

For checkpoint i , Action L1 is

$$J_{\text{L1}}(i) = \mathbb{E}_{(x_t, a_t) \sim \mathcal{D}_{\text{val}}} [\|\pi_i(x_t) - a_t\|_1]. \quad (3)$$

It is teacher-forced and has no state rollout, but directly measures imitation quality in a common action space. Open-loop regularity queries the policy along the expert trajectory and measures the second-difference curvature

$$J_{\text{smooth}}^E = \frac{1}{H-2} \sum_{k=2}^{H-1} \|a_k^E - 2a_{k-1}^E + a_{k-2}^E\|_2^2. \quad (4)$$

PLV-1 applies the same curvature statistic to actions generated recursively by $\hat{a}_k = \pi_i(\hat{x}_k)$ and $\hat{x}_{k+1} = f_\psi(\hat{x}_k, \hat{a}_k)$. The “1” denotes a recursively applied one-step transition model, not a one-step evaluation horizon. Comparing PLV-1 with open-loop smoothness tests whether modeled feedback adds information beyond action regularity.

B. Execution-matched replanning

Action-chunking policies output $A_t = \pi(x_t)$, but the deployed Diffusion Policy uses an executable offset $s = 1$ and commits

$$A_t^{\text{commit}} = A_t[s : s + m] \quad (5)$$

before replanning. We therefore compare the suffix after this slice with the time-aligned prefix of the replacement plan. The main audit uses stride $m = 8$, prediction horizon 16, observation horizon two, and absolute actions.

At replan cycle $r + 1$, boundary jump compares the last committed action with the first action in the replacement plan:

$$J_{\text{boundary}} = \frac{1}{d_a} \left\| \frac{A_{t+m}[s] - A_t[s + m - 1]}{\sigma_a} \right\|_1. \quad (6)$$

Here σ_a is the demonstration action standard deviation. Because the audit uses absolute actions, the quantity reflects

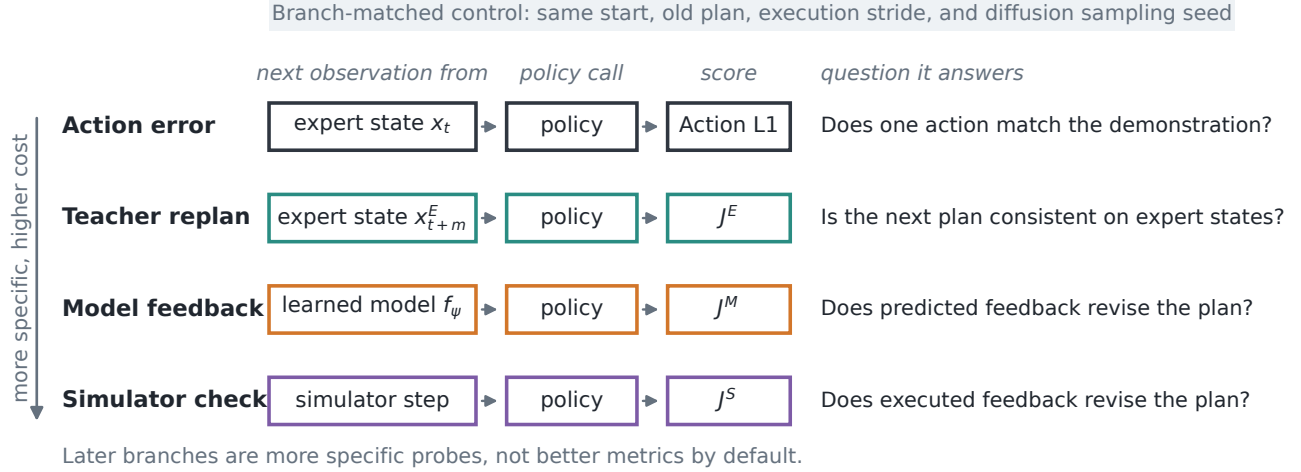


Fig. 1. Evaluation hierarchy. Every row shares the same three stages; only the source of the next observation changes. Later rows are more specific probes, not better metrics by default: each answers a narrower question at higher cost per checkpoint, under the same start, old plan, execution stride, and sampling seed.

a discontinuity in the commanded pose/action, not a jump between action deltas.

Replan overlap aligns actions that refer to the same future times. If L old-plan actions remain after the executed slice,

$$J_{\text{replan}} = \frac{1}{Ld_a} \left\| \frac{A_t[s+m : s+m+L] - A_{t+m}[s : s+L]}{\sigma_a} \right\|_1. \quad (7)$$

Large values mean that the new replan observation caused the policy to revise its future intent. This is inter-plan inconsistency, not smoothness within one committed chunk.

C. Teacher, model, and simulator branches

The same-start audit fixes the validation start, old plan, execution stride, and diffusion seed, then changes only the source of the next observation. Teacher-forced replanning uses the recorded expert state, model-induced replanning uses compact learned dynamics, and simulator-induced replanning executes the old plan in robosuite:

$$A_{t+m}^E = \pi(x_{t+m}^E), \quad (8)$$

$$\hat{x}_{t+m}^M = f_\psi^{(m)}(x_t, A_t), \quad A_{t+m}^M = \pi(\hat{x}_{t+m}^M), \quad (9)$$

$$x_{t+m}^S = \text{Env}^{(m)}(x_t, A_t), \quad A_{t+m}^S = \pi(x_{t+m}^S). \quad (10)$$

Each branch reconstructs the policy’s two-observation history from its own source; the supplement gives the per-branch construction.

Let J^E , J^M , and J^S denote the overlap inconsistency in each branch. The teacher-corrected excess terms are

$$\Delta J^M = J^M - J^E, \quad \Delta J^S = J^S - J^E. \quad (11)$$

The archive persisted these branch costs but not the replacement-plan vectors needed for direct common-seed branch contrasts. We therefore treat ΔJ as a weaker proxy, not a clean identification of state feedback; the direct estimands and archive limitation are detailed in the supplement.

Diffusion stochasticity is separated with four policy seeds into a sampling-only distance, a state-only distance under common random numbers, and a deployment-total distance that varies both; scores average starts within each seed and take the median across seeds. The supplement gives the three definitions.

The main robomimic study uses normalized low-dimensional observations x_t . We reserve $z_t = \phi(o_t)$ for a supplementary representation audit; learned latent structure has no demonstrated advantage over capacity-matched raw state. Each task-local dynamics ensemble has five two-layer MLPs of width 256 and is trained on at most 30,000 demonstration transitions for eight epochs. We call recursive one-step smoothness *PLV-I*, execution-matched model-induced replanning *PLV-M*, and use PLV only for the umbrella framework.

V. EXPERIMENTAL PROTOCOL

The primary retrospective cohort contains low-dimensional robomimic and robosuite Can, Lift, Square, Transport, and Tool Hang tasks [5], [6]. Here “primary” identifies the main analysis cohort, not a preregistered confirmation set. BC-RNN and Diffusion Policy checkpoints cover PH, MH, MG, and Paired demonstrations when available, and every checkpoint has 50 simulator rollouts. The historical execution archive contains 504 Diffusion Policy checkpoints. Four-seed controls could be recomputed for 130 retained binaries, while the teacher/model/simulator same-start audit uses all 47 locally available nonzero-epoch binaries in nine outcome-blind retained grids. Each grid contains five or six checkpoints; residual estimation is finite in six because outcomes or predictors are degenerate in the other three. These retained subsets are mechanism checks, not new complete grids. MimicGen, ACT, and RoboCasa cohorts test broader tasks, complete-grid transfer, and policy-family boundaries [7], [8]. The full cohort map,

including non-additive nested subsets and evidence grades, is provided in the supplement.

Metric direction, checkpoint inclusion, and outcome chronology are recorded before each frozen external analysis, and outcome columns are read only after scores are frozen. The strict prospective tests were preregistered but not fully analyst-blind (see limitations). The final same-start gate is explicitly post-hoc because its aggregation criterion was strengthened after launch; policy inference, dynamics training, simulator stepping, and raw metric generation were unchanged. The supplement records this deviation and the separate provenance of later breadth extensions.

Rollout success is quantized in increments of $1/50$. We propagate this measurement noise with a within-checkpoint binomial episode bootstrap while retaining the run and task hierarchy. We also condition on each observed success count and repeatedly allocate it between two 25-episode halves with the exact hypergeometric distribution. It therefore captures conditional binomial sampling noise, not variation from unarchived initial conditions or environment seeds. In addition to τ_b , we sample a $\text{Beta}(k_i + 1, 50 - k_i + 1)$ posterior for each checkpoint and score pairwise ordering only when $P(S_i > S_j) > 0.95$ or its reverse exceeds that threshold. This avoids treating statistically unresolved differences such as 0.48 versus 0.50 as certain orderings.

Analysis code, manifests, and hashed result tables are packaged with the paper source and maintained at <https://github.com/cheesss/plv-robot-imitation-learning>.

VI. RESULTS

A. RQ1: What do cheap metrics actually select?

The primary result is structured heterogeneity, not a single winning score. After separating dataset and seed runs and weighting tasks equally, Action L1 and PLV-1 smoothness reach 0.418 and 0.434, respectively. Their hierarchical intervals overlap and the task-level difference is small ($\Delta\rho = 0.016$). Selection endpoints favor Action L1: MMRV is 0.153 versus 0.160, and Regret@30 is 0.033 versus 0.045. Random Regret@30 is 0.046 with interval $[0.026, 0.064]$; open-loop smoothness reaches 0.019 and is the only reported screen below this interval.

The conditional split-half estimate is 0.500, giving full-50 reliability 0.667; a one-sided attenuation approximation raises Action-L1/PLV-1 to near 0.512/0.532 (supplement). Noise therefore explains part, not most, of the remaining selector gap.

Objective validation loss is not uniformly weak. Its within-run association with success varies across policy families and data regimes, producing a task-balanced primary mean of only 0.045. In the BC-RNN subset it has slightly lower ρ than Action L1 (0.360 versus 0.372) but better Regret@30 (0.034 versus 0.050); in matched external BC-RNN cohorts it is strongest by both criteria. The policy-family split also explains the close pooled PLV result. Across 24 Diffusion Policy runs, PLV-1 leads Action L1 on ρ , MMRV, and Regret@30 (0.556/0.093/0.006 versus 0.462/0.120/0.017),

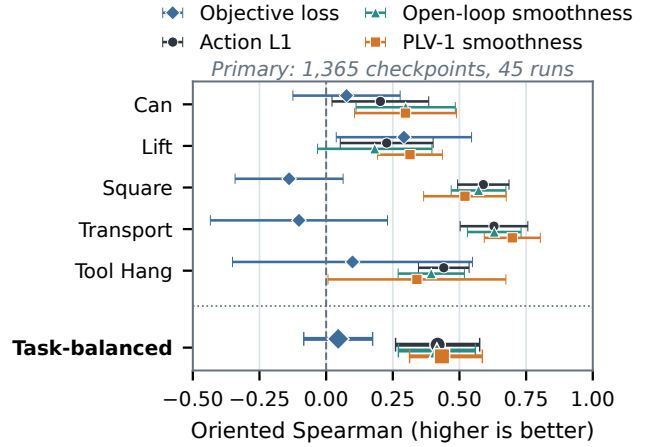


Fig. 2. Run-level ranking is heterogeneous across tasks and data regimes. Metrics are computed inside independent task-dataset-policy-seed runs; adjacent checkpoints are never treated as independent replicates. Task rows give the mean over runs with ± 1.96 standard errors; the bold row is the reported task-balanced estimate with a task-clustered bootstrap 95% interval. Costs are negated before correlation, so higher is better.

whereas across 21 BC-RNN runs the ordering reverses and open-loop smoothness leads. This reversal is a central empirical finding: selector behavior varies systematically with policy family, and averaging families can hide the regime in which a temporal diagnostic is informative (supplement).

Non-discriminative pools explain part of the instability: a metric cannot recover a meaningful order when nearly every checkpoint succeeds or nearly every checkpoint fails. We therefore report both all-run and prespecified outcome-resolving summaries, defined in the supplement. The posterior-resolvable pair and score-normalization audits give the same qualitative conclusion. This exposes a second central finding. Across metrics, most rank signal separates broken checkpoints from working ones: success rises sharply from the worst to the middle score quintile but changes little across the top 60%. A metric can therefore rank the full training trajectory well while adding little to the final checkpoint decision.

B. RQ2: What does learned feedback add beyond open-loop regularity?

The dynamics-free comparison asks whether a learned policy loop contributes more than action regularity. Open-loop smoothness reaches 0.415 task-balanced run-level Spearman and lower Regret@30 (0.019) than PLV-1 smoothness (0.434 and 0.045). The task-level gaps are much smaller than the design’s detectable effects, and exact paired tests do not resolve them (supplement). The signals are related but not identical: their mean pairwise within-run ρ is 0.53–0.66, the first rank component explains 73.9% of variance, and PLV-1 retains partial rank association 0.183 after Action L1 and open-loop smoothness. Most importantly, this residual signal is regime-dependent. On Diffusion Policy, PLV-1 exceeds open-loop smoothness by 0.138, whereas no pooled selection-regret gain

TABLE I

PRIMARY RETROSPECTIVE TASK-BALANCED RUN-LEVEL RANKING (MAIN ANALYSIS, NOT PREREGISTERED CONFIRMATION). 1,365 CHECKPOINTS, 45 TASK-DATASET-POLICY-SEED RUNS, FIVE TASKS. CORRELATIONS ARE COMPUTED INSIDE EACH RUN AND WEIGHTED EQUALLY ACROSS THE FIVE TASK MEANS; BRACKETS ARE TASK-CLUSTERED HIERARCHICAL-BOOTSTRAP 95% INTERVALS. RESOLVING ρ APPLIES THE PRESPECIFIED OUTCOME-RESOLVING FILTER DEFINED IN THE SUPPLEMENT. COSTS ARE NEGATED ONCE BEFORE CORRELATION, SO HIGHER ρ IS BETTER; REGRET@30 (30% RETAINED BUDGET) AND MMRV ARE LOWER-BETTER.

Metric	All-run $\rho \uparrow$ [CI]	Regret@30 $\downarrow$ [CI]	MMRV $\downarrow$	Resolving $\rho \uparrow$
Action L1	0.418 [0.246, 0.579]	0.033 [0.009, 0.069]	0.153	0.334
Open-loop smoothness	0.415 [0.252, 0.569]	0.019 [0.003, 0.049]	0.166	0.353
PLV-1 smoothness	0.434 [0.283, 0.592]	0.045 [0.002, 0.130]	0.160	0.434
PLV-1 uncertainty+smooth	0.425 [0.300, 0.553]	0.041 [0.005, 0.108]	0.161	0.401
Objective validation loss	0.045 [-0.134, 0.231]	0.060 [0.023, 0.105]	0.290	-0.018

is established. Learned feedback therefore adds a conditional policy-family signal, not a uniformly better scalar.

C. RQ3: What survives execution-matched controls?

Execution-matched replanning is compared with a teacher-forced expert-state branch, alongside same-observation sampling inconsistency so that diffusion variance is not mislabeled as state feedback.

The four-seed audit covers 130 retained checkpoint binaries (Table II, Panel B). Its task-balanced paired difference between the model-induced and teacher-forced scores is 0.019 across 5 tasks (two-sided task-level Wilcoxon $p = 1.000$), and the model-induced residual after Action L1 and teacher-forced replan is 0.126 over 9 eligible runs. The teacher-forced and learned-loop replan scores are statistically indistinguishable at this scale. The evidence supports replanning consistency, but does not isolate learned dynamics as its source.

Execution matching produces the paper’s clearest mechanism finding. The one-step probe revises its action after every modeled transition and makes curvature appear strong; after committing eight actions, full-chunk smoothness falls to 0.339. In the historical task-pooled hard Diffusion Policy discovery archive, boundary jump and replan overlap instead reach 0.730 and 0.707, compared with 0.512 for Action L1. The useful signal is therefore not generic within-chunk smoothness but disagreement where an executed plan is replaced. This result is descriptive because it was found during metric development.

The 130-binary retained control sharpens the interpretation (Table II, Panel B; Fig. 3b, filled markers). Its task-balanced estimates do not reproduce the historical hard-task ordering, while the model-induced residual is defined for only nine of its 24 runs. The discovery signal therefore belongs to a specific execution-boundary regime; it cannot be attributed to learned dynamics alone or promoted to automatic reranking authority.

Common random numbers isolate state-induced revision from diffusion sampling noise, so a high sampling-only score is a warning about policy stochasticity rather than proof of closed-loop state sensitivity.

a) *Simulator-reference diagnostic*: Demonstration prediction is accurate at the current budget (normalized-state MSE 0.0017 at one step and 0.0045 over five steps), yet policy-induced drift is sharp. On the same 47 checkpoints final MSE reaches 3.135 after 16 actions. Neither shortening the stride

nor widening the model recovers the intended oriented ranking: width 512 lowers it to 1.309 while making oriented model-replan ρ more adverse, from -0.125 to -0.211 (paired task change -0.085 , exact $p = 0.625$; sweep in the supplement). Raw model-simulator ordering is 0.519 but teacher-corrected excess agreement is 0.096; simulator J^S itself has an adverse oriented $\rho = -0.212$. This control establishes a useful distinction: transition fidelity, model-simulator agreement, and success ordering are separate requirements. Extrapolation is real, but improving prediction alone cannot make replan inconsistency a universal selector.

The post-hoc teacher-corrected gate fails: the model excess score has $\rho = 0.068$, residual -0.309 after Action L1 and teacher replan in six eligible runs, and descriptive Regret@30 0.017 versus 0.031 for Action L1 (two improve, two tie; exact $p = 0.500$). This is not evidence for automatic reranking.

The task-balanced model-minus-teacher replan correlation difference is 0.163 with a task-cluster bootstrap 95% interval $[-0.131, 0.542]$. The interval is wide and crosses zero; the same-start audit therefore does not establish a learned-feedback gain.

D. RQ4: Where does the effect transfer?

The external program maps the boundary of the finding; the complete evidence matrix is in the supplement. An action-chunking breadth cohort favors open-loop smoothness (0.829) over both Action L1 and PLV-1. Selected MimicGen candidates show a positive frozen temporal result, whereas complete BC-RNN grids, unseen ACT checkpoints, and six strict RoboCasa cohorts do not reproduce a general PLV advantage. The strongest prospective test is also informative: in two preregistered cohorts, 126 scores were frozen before 6,300 rollout episodes were collected per cohort. The selector missed both promotion gates, with Regret@30 changing from 0.007 to 0.017 on MimicGen domain variants and from 0.047 to 0.050 on the unseen bimanual family. A candidate-only cohort with three successes in 2,000 episodes is non-discriminative and cannot adjudicate either selector.

These tests turn an initially broad claim into a precise one. The PLV variants evaluated here are not a family-general scalar, and validation reliability is explicitly regime-dependent. At the same time, execution-matched replan overlap has not been tested on every external family. The evidence supports

TABLE II

EXECUTION-SEMANTICS AUDIT; NOT A GLOBAL SELECTION TABLE. PANEL A: TASK-POOLED HARD-DP DISCOVERY ARCHIVE (TWO TASK ESTIMATES; DISCOVERY-ONLY).
 PANEL B: MECHANISM AUDIT OVER 24 RETAINED-BINARY RUNS (130 BINARIES); RESIDUAL ρ REMOVES ACTION L1 AND TEACHER-FORCED REPLAN FIRST AND IS
 DEFINED FOR 9 RUNS. REPLAN COSTS ARE NEGATED FIRST, SO A HIGHER ORIENTED ρ IS BETTER. THE PANELS USE DIFFERENT UNITS, ARE NEVER POOLED, AND A DASH
 MARKS AN UNDEFINED QUANTITY.

Metric	All DP $\rho \uparrow$	Hard DP $\rho \uparrow$	Can control $\rho \uparrow$	Residual $\rho \uparrow$
<i>A. Historical hard-DP discovery (2 task-pooled estimates)</i>				
Action L1	–	0.512	–	–
One-step PLV smoothness	–	0.745	–	–
Chunk PLV smoothness	–	0.339	–	–
Boundary jump	–	0.730	–	–
Replan overlap	–	0.707	–	–
<i>B. Four-seed retained-binary control (24 task/dataset/seed runs)</i>				
Four-seed Action L1	0.434	0.520	0.540	–
Teacher-forced replan	0.386	0.384	0.423	–
Sampling-only inconsistency	0.327	0.363	0.311	–
Model-induced replan	0.405	0.328	0.264	0.126
Model+sampling replan	0.337	0.316	0.313	0.066

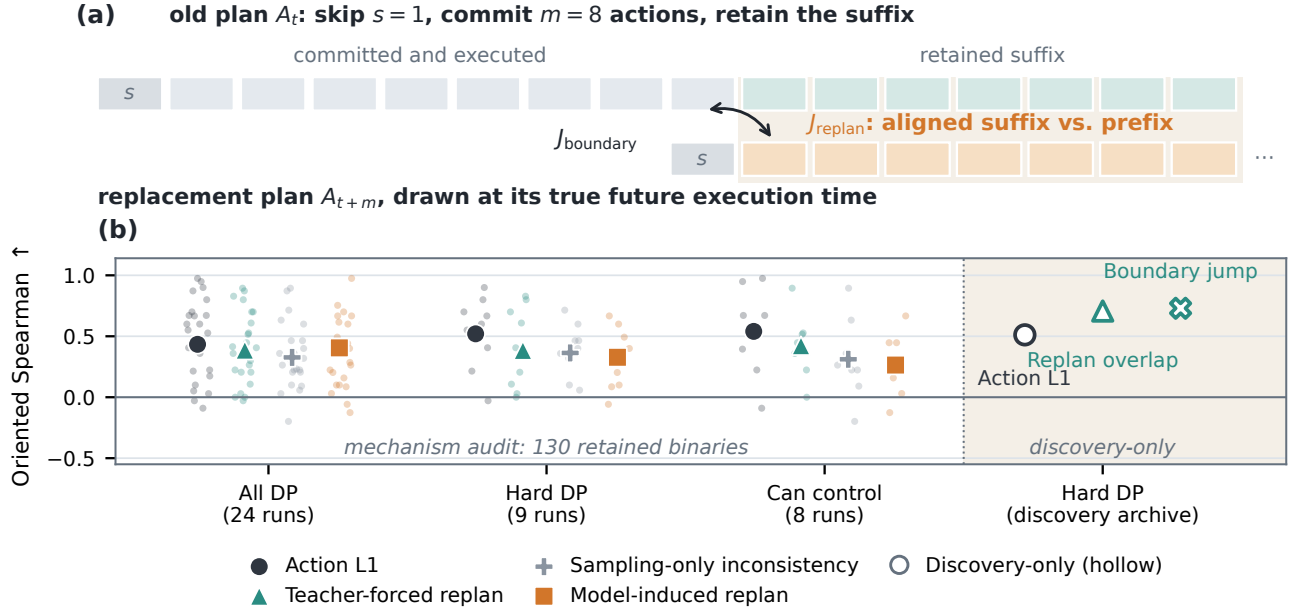


Fig. 3. Replanning mechanism and ranking result. (a) The old plan commits $m = 8$ actions after the skipped offset $s = 1$; the replacement is drawn at its true future execution time, so J_{replan} compares the retained suffix with the aligned prefix and J_{boundary} compares the last committed with the first replacement action. (b) Filled markers are the retained-binary mechanism audit (small dots: its runs; large markers: task-balanced mean); hollow markers on the shaded band are the task-pooled discovery archive and are never combined with it. Replan costs are negated before correlation; higher oriented Spearman is better.

temporal diagnosis inside the known action-chunking regime, with fresh calibration or abstention elsewhere.

VII. IMPLICATIONS FOR SCREENING

The practical output of this study is an evidence hierarchy rather than another universal scalar. Action L1 and open-loop regularity provide low-cost evidence of imitation quality and temporal regularity; teacher-forced replanning asks whether future intent is stable; model and simulator branches test whether feedback changes that intent. These questions should not be collapsed into one ranking. When the cheap screens agree, they provide a defensible first pass. When a competent action-chunking checkpoint shows temporal disagreement, the

evidence identifies a candidate for targeted rollout rather than silently overriding the cheap ranking.

This cheap-screen-diagnose-abstain interpretation is supported by the comparative evidence, but its use as an allocation policy remains future work: the one frozen prospective allocation attempt was non-discriminative. Applying learned dynamics to very poor checkpoints may drive imagined trajectories into a shared extrapolation region where scores saturate, but the retained same-start audit is too small to establish that mechanism. Tail aggregation does not fix the issue in the archived execution pool, and we omit an “equivalent rollout” cost curve because archived runtimes lack a common hardware

TABLE III

SAME-START REPLAN AUDIT USING THE ARCHIVED TEACHER-CORRECTED EXCESS PROXY; EVIDENCE GRADE: MECHANISM AUDIT, NOT A DEPLOYMENT SELECTOR. THE SAMPLE UNIT IS THE RUN: ALL LOCALLY AVAILABLE NONZERO EPOCHS IN NINE OUTCOME-BLIND RETAINED GRIDS (47 CHECKPOINTS, FOUR TASKS), EACH BRANCH SHARING THE SAME STARTS, EXECUTION STRIDE, AND FOUR DIFFUSION SEEDS. RISK COSTS ARE NEGATED BEFORE SUCCESS CORRELATION, SO POSITIVE VALUES INDICATE THE INTENDED SELECTOR DIRECTION; EXCESS AGREEMENT INSTEAD CORRELATES ΔJ^M WITH ΔJ^S . DIRECT REPLACEMENT-PLAN CONTRASTS WERE NOT PERSISTED HERE.

Task	Runs	Action L1	J^E	J^M	J^S	ΔJ^M	ΔJ^S	Agree. ρ	Residual ρ
Can	2	-0.433	-0.712	0.027	-0.824	0.712	-0.499	-0.221	–
Square	2	-0.600	-0.364	-0.412	-0.156	-0.469	-0.053	0.300	-0.399
Tool Hang	2	-0.026	-0.020	-0.233	-0.134	0.006	-0.405	0.071	-0.095
Transport	3	-0.191	-0.056	0.118	0.268	0.021	0.274	0.233	-0.432
Task-balanced	9	-0.312	-0.288	-0.125	-0.212	0.068	-0.171	0.096	-0.309

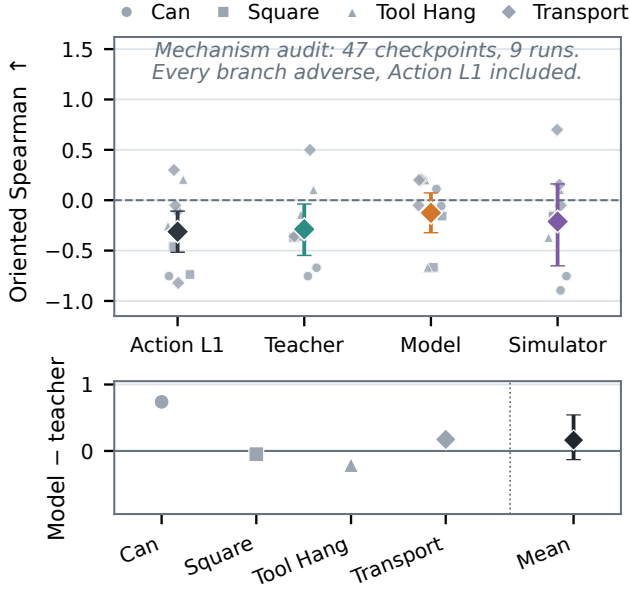


Fig. 4. Same-start audit: a control experiment, not a deployment selector. Gray markers are the run-level estimates with shape giving the task; colored diamonds are task-balanced means with task-cluster bootstrap 95% intervals. Replan costs are negated before correlation; higher oriented Spearman is better. The lower panel shows the paired model-minus-teacher difference by task; its interval spans zero.

benchmark (supplement).

VIII. DISCUSSION AND LIMITATIONS

a) Policy family is part of the evaluation problem:

The BC-RNN-Diffusion-Policy reversal is not a nuisance to average away. The pattern is consistent with BC-RNN checkpoints being well described by objective or action-space fit, while action chunking introduces commitment and replanning boundaries that pointwise error cannot represent. Learned dynamics is therefore most defensible as a research probe in outcome-varying action-chunking pools where Action L1 and teacher-forced replanning disagree. Its value is the additional question it asks, not automatic authority over the final ranking.

b) Why ranking and selection disagree:

All metrics obtain much of their rank correlation by separating broken checkpoints from working ones, while the top of each pool is

TABLE IV

DIAGNOSTIC GATES. THE FIRST THREE ROWS ARE PRIMARY (45 RUNS, 1,365 CHECKPOINTS); THE DYNAMICS ROWS ARE THE MECHANISM AUDIT (NINE RUNS, 47 CHECKPOINTS). RELIABILITY AND ρ ARE HIGHER-BETTER; REGRET AND MSE ARE LOWER-BETTER, AS MARKED. SPLIT-HALF RELIABILITY IS CONDITIONAL ON THE 50 OBSERVED BINARY OUTCOMES. THE DYNAMICS ROWS USE NORMALIZED LOW-DIMENSIONAL STATE; ONE/FIVE-STEP DEMONSTRATION AND 16-STEP SAME-START VALUES ARE DIFFERENT FROZEN PROTOCOLS.

Check	Result Reading
25/25 split $\rho \uparrow$	0.500 [0.472, 0.529] substantial noise
Full-50 reliability $\uparrow$	0.667 noise is not the sole limit
Random Regret@30 $\downarrow$	0.046 Action L1 0.033, PLV-1 0.045
Demo dynamics MSE $\downarrow$	0.0017 / 0.0045 1 / 5 steps
Same-start final MSE $\downarrow$	3.135 16-step drift
Simulator $J^S \rho \uparrow$	-0.212 adverse ordering

comparatively flat. Inside the retained top 30%, the estimated true-ability spread is only 0.042 and is concentrated on harder tasks (0.083 on Tool Hang versus 0.008 on Can). This narrow, uneven headroom explains why global ρ does not guarantee low selection regret and why conflicting top-of-pool scores should trigger a rollout.

c) *Prediction quality is not decision quality:* Mechanism studies point to the same boundary:

$$\text{transition calibration} \Rightarrow \text{checkpoint ordering.} \quad (12)$$

A probabilistic ensemble improves multi-step uncertainty-error calibration without a comparable ranking gain: aleatoric uncertainty can predict transition error while adding policy-independent noise to policy ordering. Learned progress and latent paired-excess variants likewise do not transfer reliably. An evaluator must preserve decision-relevant ordering, not merely fit transitions or local semantics.

d) *Limitations:* The strict external evaluations were pre-registered but not fully analyst-blind: code, budgets, and gates were frozen, yet routine integrity checks exposed checkpoint filename scores and partial rewards before aggregate analysis. No per-episode success labels or confirmatory reports were exposed, and the analysis was unchanged (supplement). The same-start mechanism audit is availability-limited, and its final aggregation gate is post-hoc.

Policy family, stride, recurrence, and difficulty covary, so the hard action-chunking result is not causal. Success counts are discrete, and the 25/25 audit is exact only conditional

on archived counts. The evidence is low-dimensional and simulation-only. Finally, negative model claims apply only to the tested five-member MLP ensembles, 30,000-transition budget, and eight training epochs; widths 256 and 512 do not define a scaling law.

IX. CONCLUSION

This study provides a controlled account of what offline checkpoint metrics can and cannot decide in robot imitation learning. Three findings are consistent across the analysis. First, policy family matters: objective and action-space metrics remain strong for BC-RNN, while PLV-1 carries more ranking signal on Diffusion Policy. Second, global ranking and final selection are different problems because most metrics separate broken checkpoints well but face little resolvable headroom near the top. Third, accurate transition prediction does not guarantee correct checkpoint ordering.

These findings give PLV a specific role. It is not a universal replacement for Action L1; it is a temporal diagnostic for action-chunking regimes. Execution-matched boundary and overlap inconsistency reveal risk in the hard Diffusion Policy discovery pool that pointwise action error misses, while matched controls identify where that evidence stops. The resulting principle is simple: use cheap metrics to screen competence, use replanning diagnostics to expose disagreement, and request rollout when the offline evidence conflicts. A frozen allocation trial on a complete, outcome-resolving action-chunking grid is the next step from this empirical result to a deployment protocol.

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